

Fake News Detection Using Deep Learning

Deep Learning (CSCE 5218)

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Introduction

This project focuses on building a system that can automatically identify whether a news article is real or fake using machine learning or deep learning techniques. With the increasing spread of misinformation on online platforms and social media, it has become important to develop tools that can help detect unreliable or misleading news content. This project aims to analyze textual information from news articles and learn patterns that distinguish genuine news from fabricated information.

The model will be trained on a dataset containing labeled examples of real and fake news articles. By learning from these examples, the model can classify new articles and predict their authenticity. The overall goal of the project is to explore how data-driven methods can be used to combat misinformation and improve the reliability of information shared online.

Problem Statement

The problem addressed in this project is identifying whether a news article contains real information or fake or misleading content. The input to the system is the textual content of a news article, which may include the headline and the body of the article. The desired output is a classification that indicates whether the article is real or fake. The challenge arises because fake news is often written in a way that appears credible, making it difficult for readers to distinguish it from legitimate news sources.

For example, an input could be a news headline and article text such as “Scientists confirm water discovered on Mars by NASA mission.” The expected output for this input would be real news if the information is factual and verified. On the other hand, an article

with a headline like “Government secretly controlling weather through hidden satellites” may be labeled as fake news if it contains fabricated or misleading claims. The goal of the problem is to correctly determine the authenticity of news articles based solely on the information provided in the text.

Dataset

The dataset used for this project is the Fake News Detection Dataset available on Kaggle. It contains news articles labeled as either real or fake, which makes it suitable for studying the problem of misinformation detection. The dataset includes features such as the title, text (article content), subject, and date of publication. In total, the dataset contains 44,898 news articles, consisting of 23,481 fake news articles and 21,417 real news articles.

For this project, the dataset will be divided into three subsets: training, validation, and test sets. The training set will contain 31,429 instances (70%), the validation set will contain 4,490 instances (10%), and the test set will contain 8,979 instances (20%). The task is a binary classification problem, meaning the system predicts one of two classes: fake news or real news. The dataset is publicly available and can be downloaded from Kaggle: <https://www.kaggle.com/datasets/emineyetm/fake-news-detection-datasets>.

Baselines and Evaluation Metrics

In order to test how effective the deep learning models are, they will be compared to the baseline methods. The baseline methods are required in order to test whether the proposed models are actually learning anything or are just performing as well as they are due to their complexity. The baseline methods that we will use for this purpose are:

Most Frequent baseline, which will use the most common class present in the training set

Uniform baseline, which will randomly classify the data points into classes with equal probability

Stratified baseline, which will classify the data points based on the class distribution present in the training set.

The performance of the models will be tested based on their accuracy, precision, recall, and F1 score. Additionally, the confusion matrix could be used in the evaluation of the performance of the models, as it is applicable to the binary classification problem of fake news detection.

Experiments and Expected Results

In this project, we plan to experiment with several deep learning architectures for the task of fake news classification. Based on my initial exploration of the dataset and existing implementations, the models I intend to compare include LSTM, CNN, Bidirectional LSTM, GRU, and an attention-based architecture. These models were selected because they are commonly used for text classification tasks and are capable of learning meaningful patterns from sequential text data.

The experiments will be conducted using the same processed dataset so that all models can be compared fairly. The text input will first be cleaned, tokenized, converted into numerical sequences, and padded to a fixed length before being fed into the networks. We will train each model on the training set and evaluate them using the same testing procedure. In addition to comparing deep learning models, I will also include simple baseline classifiers to determine whether the neural network models provide significant improvement over basic approaches.

From initial exploration, recurrent neural models such as LSTM, Bidirectional LSTM, and GRU appear to be strong choices for this problem because they can capture sequential relationships in text. The CNN model may also perform well because convolutional filters can detect important local text patterns. On the other hand, more complex architectures such as ResNet-style models may not be as effective for this type of text representation. Therefore, we expect the recurrent models, especially GRU and Bidirectional LSTM, to achieve the strongest performance among the architectures tested.

The project's final outcome will include a comparative analysis of the models and their classification performance, as well as the identified strengths and weaknesses of each model. We will not only report evaluation metrics, but we will also attempt to explain the results, focusing on the models that excel at classifying fake news headlines and elucidating the reasons behind their superior performance.